

CHANAKYA

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Summary

Data-driven AI/ML engineer with hands-on experience in developing and deploying machine learning models for real-time applications. Proficient in Python, SQL, NLP (including classification, retrieval, and regex), and major AI frameworks such as PyTorch and TensorFlow. Experienced in applying ML solutions in production environments, including automation systems and real-time object detection. Adept at cross-functional collaboration, model evaluation, and prompt engineering, with a strong foundation in computer science and a deep interest in GenAI applications.

Education

Texas A&M University, Corpus Christi

2023-2025

Master's degree in Computer Science

Skills

Technical: Python, SQL | Machine Learning | Neural Networks | Prompt Engineering | Natural Language Processing | Object Oriented Programming | Database Management and data Analysis | MATLAB and Simulink | Automation and Scripting

Other: Communication, Leadership and Team Collaboration | Time Management and Organization | Problem Solving

Experience

Tata Elxsi, India

July 2022 – August 2023

MBD Engineer

- Designed and developed models based on client requirements using MATLAB and Simulink.
- Automated data handling processes by writing Python scripts, reducing manual workload by 40%.
- Collaborated with cross-functional teams to ensure quality and timely delivery of scalable solutions.
- Received recognition for innovative ideas and optimizing workflows, saving project sprints.

Tata Elxsi, India

February 2022 - June 2022

Intern

- Received hands-on training in MATLAB and Simulink directly from MathWorks engineers, focusing on automotive model-based design principles.
- Developed control logic models for an advanced Power Window System, a premium feature used in high-end vehicles.
- Strengthened foundational skills in embedded system modeling, component interaction, and system-level simulations.

Projects

Real-Time Hazard Detection System for Children's Play Area Safety

- Developed a real-time computer vision system to enhance safety in children's play areas by identifying hazardous objects.
- Implemented object detection algorithms (YOLO, SSD, Faster R-CNN) for accurate and efficient detection and alerting caregivers.
- Focused on improving detection accuracy, processing speed, and minimizing false alarms in dynamic environments.
- Conducted data annotation, augmentation, and applied transfer learning to ensure robust and reliable performance.

Enhancing Deepfake Detection with Advanced Neural Networks

- Built an ensemble deepfake detection model (CNN, LSTM, ViT, FFT) with 95% accuracy on Celeb-DF and DFDC datasets.
- Designed evaluation pipeline (F1, ROC-AUC) and performed detailed error analysis for model validation.
- Contributed to real-time, explainable ML deployment with Grad-CAM visualizations and GPU-optimized inference

CrossConnect: Seamless Cross-Border Shopping Platform

- Designed and developed CrossConnect, a platform enabling seamless cross-border shopping by connecting users in the USA with verified local agents in India for authentic Indian products.
- Implemented key features including real-time agent communication, transparent pricing, secure transactions, and fast delivery to ensure a user-friendly and trustworthy experience.

Automation of HVAC components for cars

- Designed and developed MATLAB Simulink models for stepper motor control to automate HVAC functionality in cars, aligning with client specifications.
- Tested and validated the Simulink models, generated production-ready code, and integrated it with other vehicle components.
- Conducted system testing to ensure premium functionality, deploying the solution into vehicles for real-world applications.

Honors

Innovation Award

at Tata Elxsi for implementing Python-based automation that reduced project timelines significantly.

- Commended for initiative in budget optimization during on-campus employment.